

Bluesky: A Collaborative Toolbox

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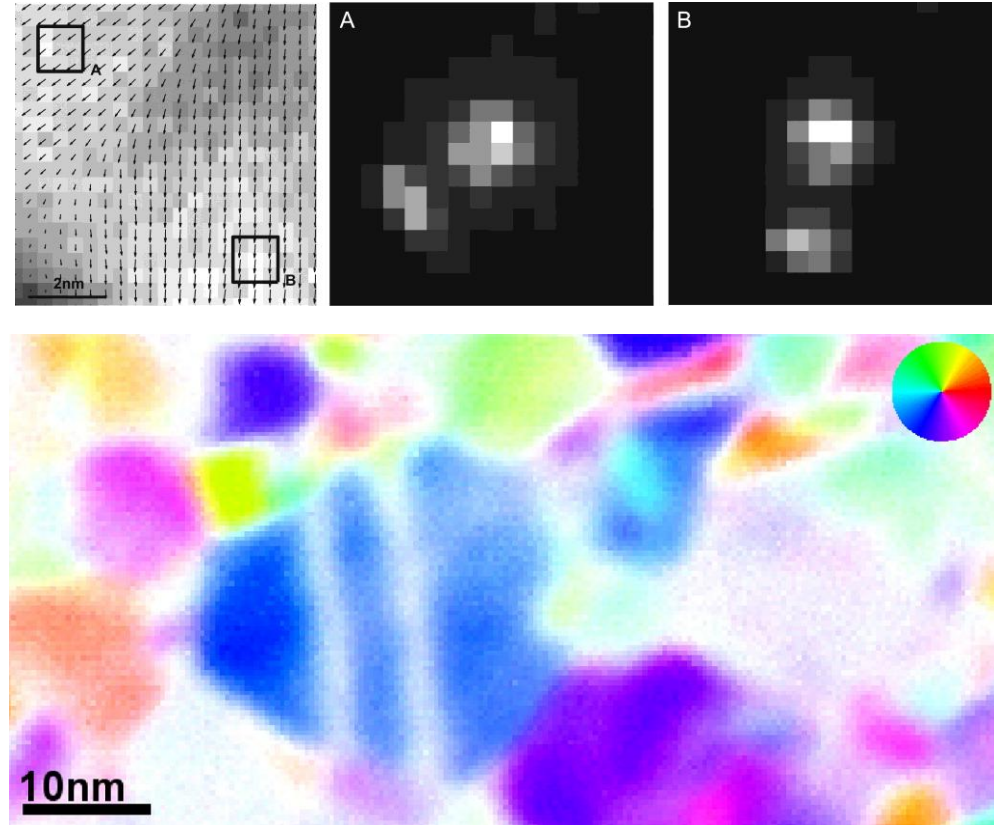
Summer School on ML for EM, June 2026

Who Am I

- Ph.D. in Experimental Condensed Matter Physics (UChicago)
- 12 years in a Research Software Engineering role at NSLS-II
- Started the *Bluesky* project, a green-field (blue sky...) open-source Python approach to data acquisition and data access experimental science, with collaborators Dan Allan (NSLS-II) and Ken Lauer (NSLS-II, LCLS, industry)
- *Bluesky* has grown within NSLS-II and around the world
- Project Lead for Matplotlib, core contributor to h5py

Electron Microscopy Connection

- Worked with Sol Gruner and David Muller on first demonstration of EMPAD in a STEM
- Showed it worked and crystal segmentation



T.A. Caswell, P. Ercius, M.W. Tate, A. Ercan, S.M. Gruner, D.A. Muller, A high-speed area detector for novel imaging techniques in a scanning transmission electron microscope, *Ultramicroscopy*, Volume 109, Issue 4, 2009, Pages 304-311, ISSN 304-3991, <https://doi.org/10.1016/j.ultramic.2008.11.023>.

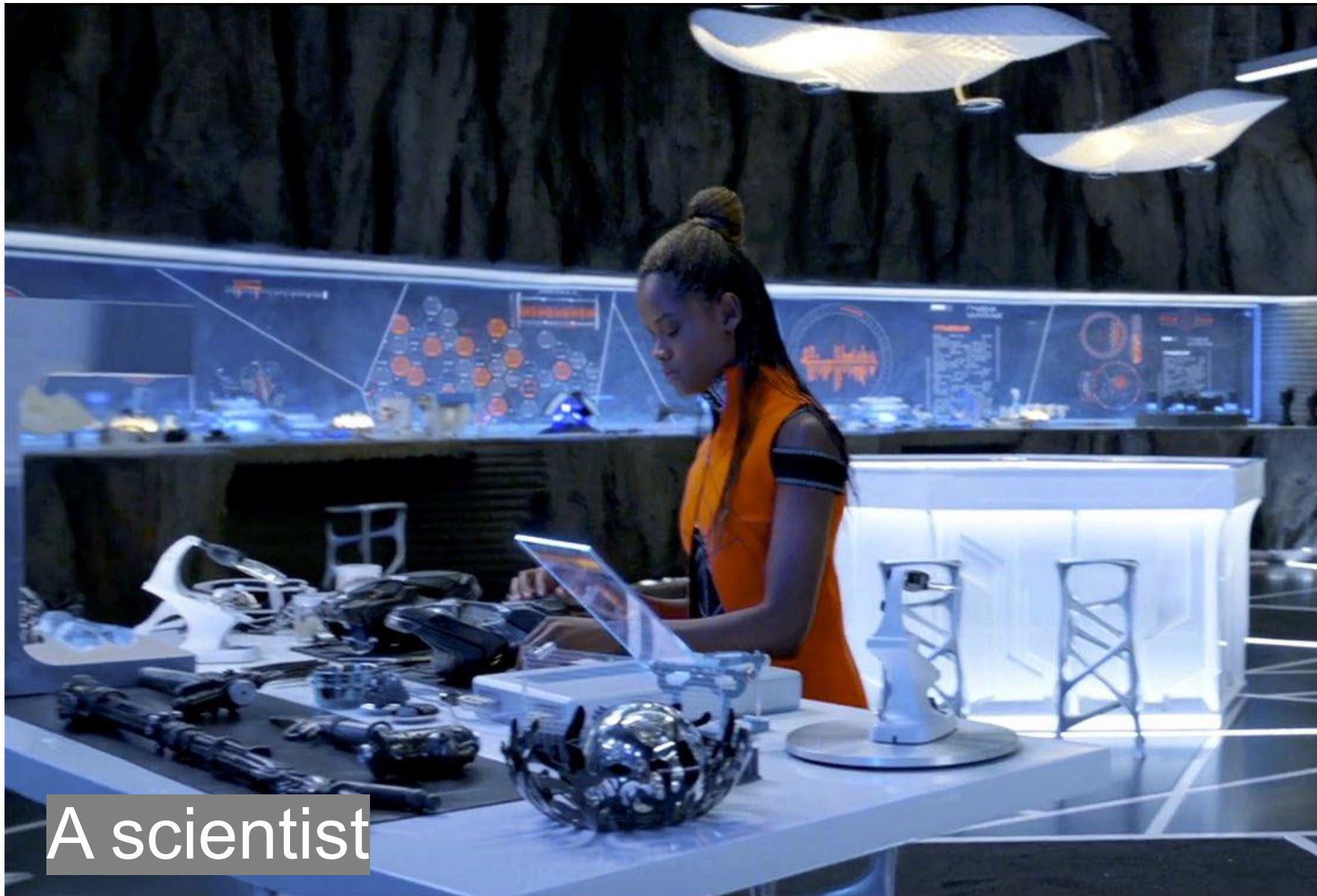
Part 1: The Duct Tape Budget



"The handyman's secret weapon"

Data
Acquisition
Challenge:

Deliver reliable
solutions that
can field new
requirements
all the time.



A scientist

Scientists are good at inventing rapidly evolving requirements.



A facility user

Microscope users are good at it too.

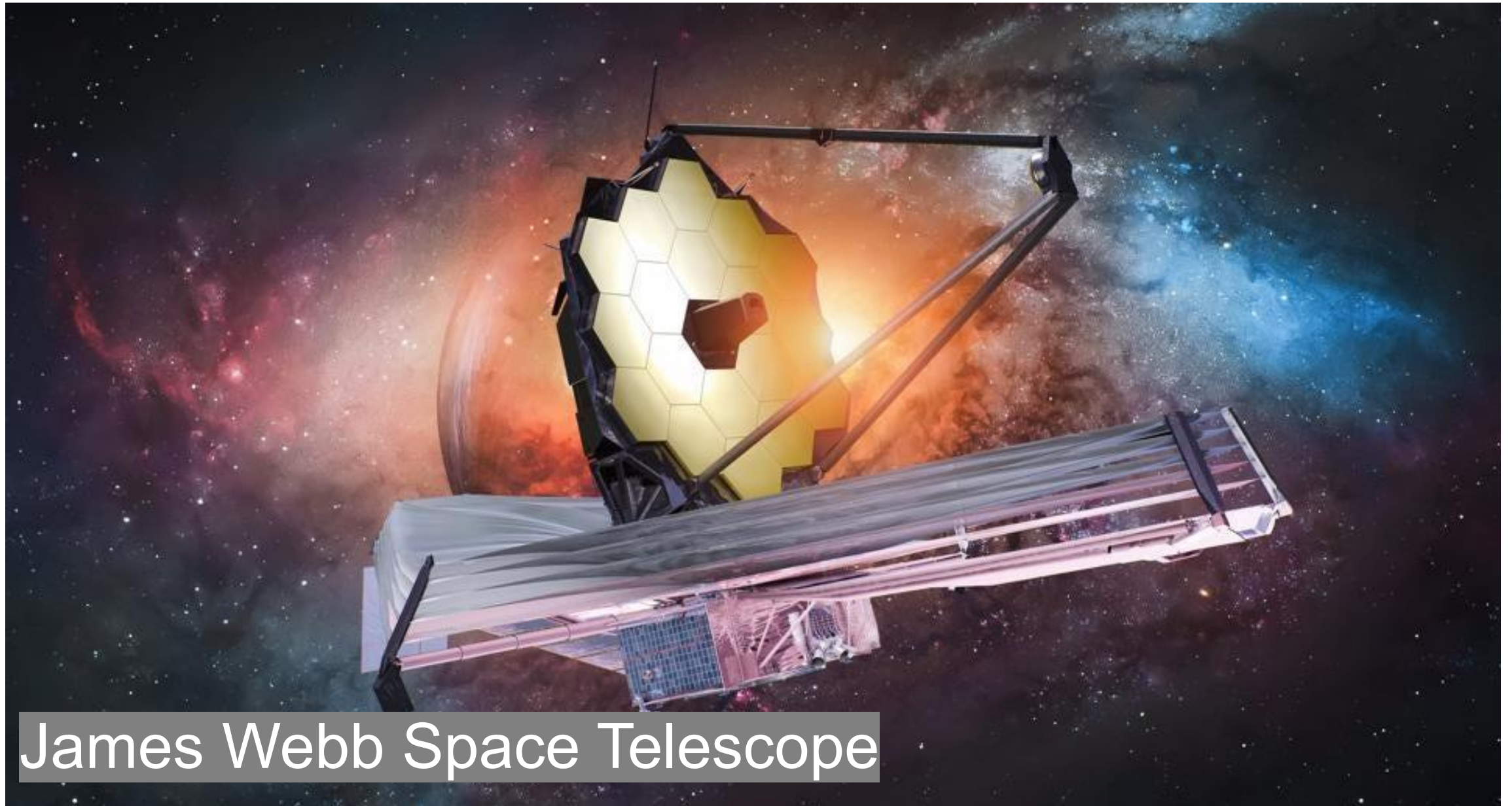
That creativity drives the science!



A controls system, c. 1970

National Synchrotron Light Source II

In 2023, NSLS-II now has 1000 networked devices on the beamlines and 2300 in the accelerator.

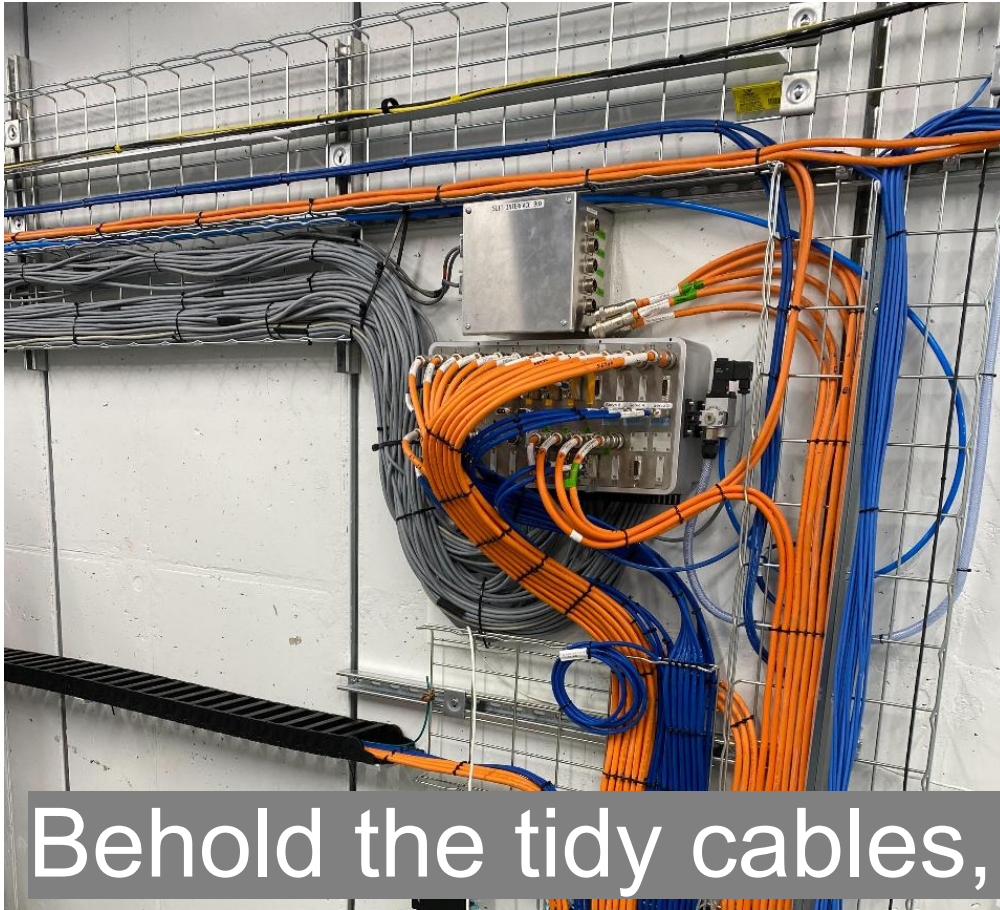


James Webb Space Telescope

How do we cope with this challenge?

1. Sound planning and engineering
2. Abstraction
3. Collaboration

Tool #1: Set ourselves up for success with sound engineering



Behold the tidy cables, and know that all will be well

Large projects are inherently complex and require many kinds of expertise



National Synchrotron Light Source II



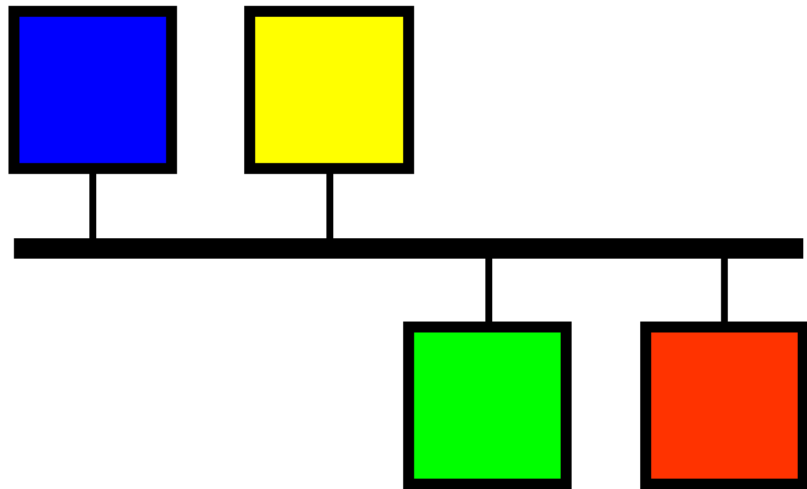
Slide adapted from Marcus Hanwell, inspired by slides from Charles Ferenbaugh, LANL

Tool #2: Use abstractions

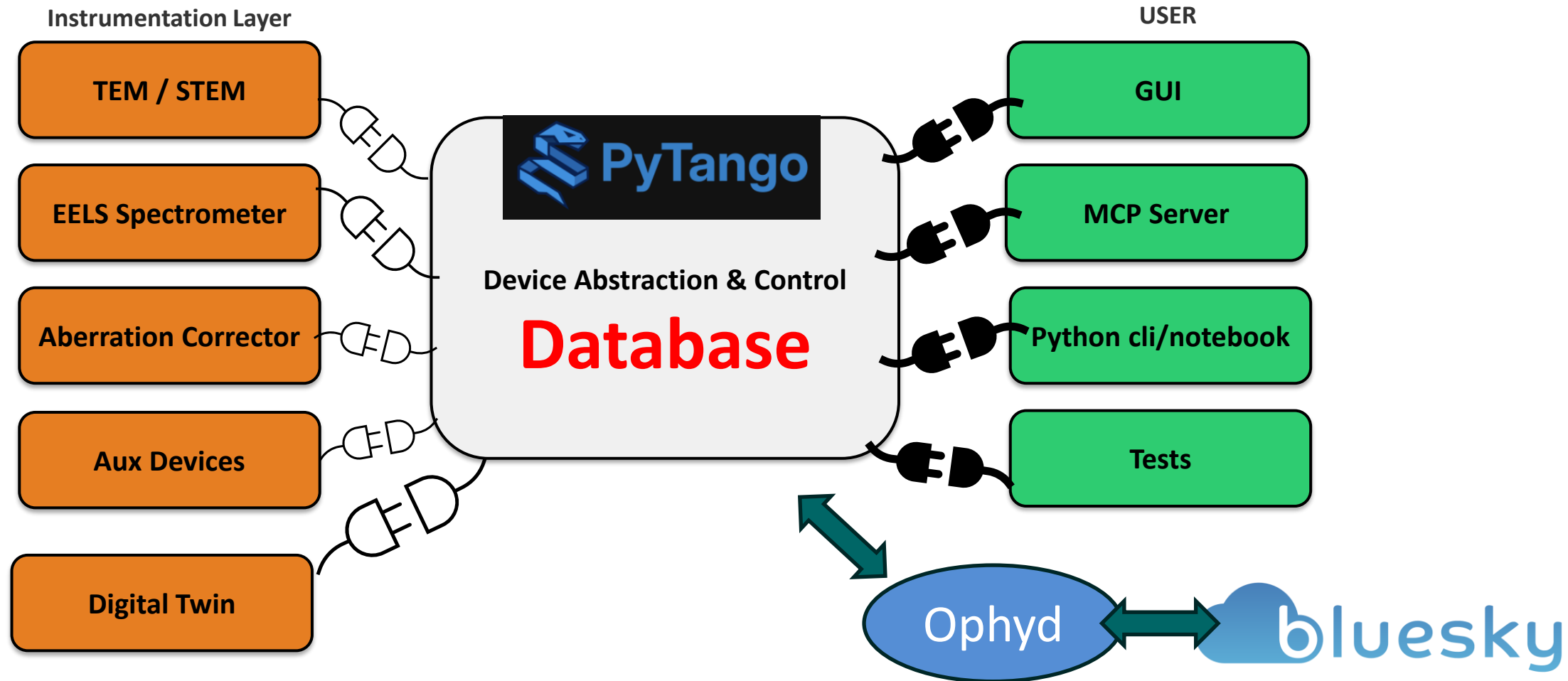


EPICS: Works for space lasers and synchrotrons

EPICS



Architecture for unified control of electron microscopy hardware using PyTango



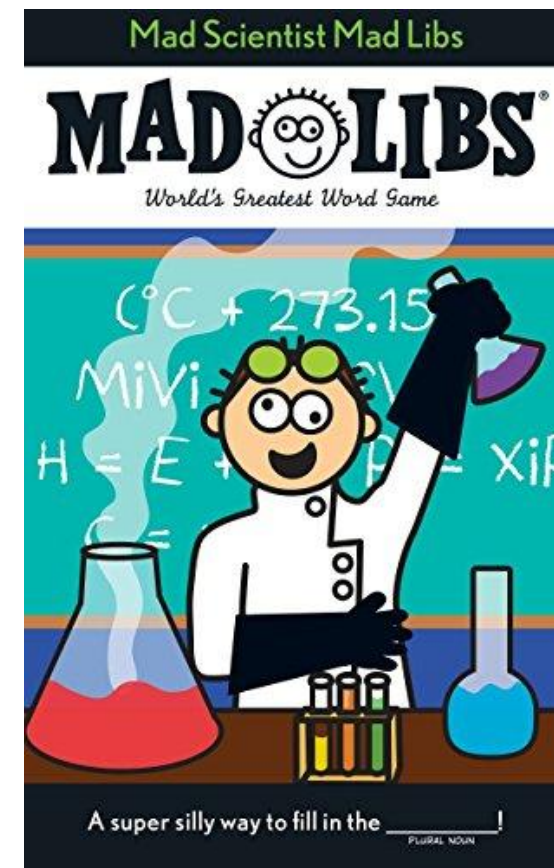
EPICS/Tango smoothes over the details of making devices do what we want

- Which computer can control this device?
- What is its current value?
- Set it to this new value (and tell me when it gets there).
- Send me updates when anything changes.

Keep device-specific details from overwhelming us

Bluesky is "Mad Libs" for science experiments

Scan the sample stage in a grid pattern over some range with some resolution while taking measurements with these sensors at each position. Record that the sample under study was Sample X and that the mood at the beamline that day was hopeful.



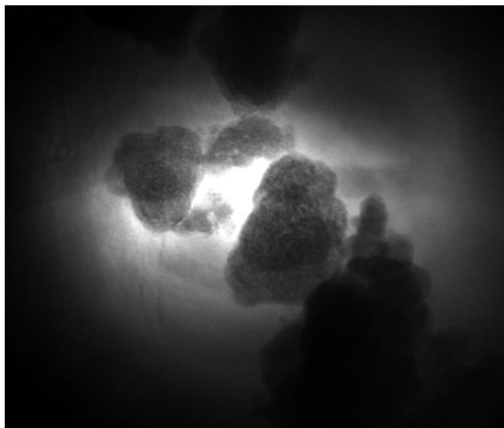
Tiled delivers data the same, whether it came from Bluesky or something else

TILED BROWSE

[Top](#) / [fxi](#) / [raw](#) / [1b0b4d73-6d87-43ab-8d62-ed035c51b9b4](#) / [primary](#) / [data](#) / [Andor_image](#)

[VIEW](#) [DOWNLOAD](#) [METADATA](#) [DETAIL](#)

Andor_image



Choose a planar cut through this 4-dimensional array.

This large array has been downsampled by a factor of 2. Use the "Download" tab to access a full-resolution image.

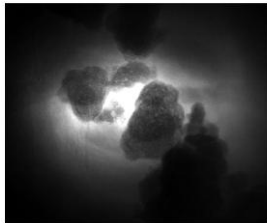
0 22
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[VIEW](#) [DOWNLOAD](#) [METADATA](#) [DETAIL](#)

Andor_image



Choose a planar cut through this 4-dimensional array.

This large array has been downsampled by a factor of 3. Use the "Download" tab to access a full-resolution image.

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TILED BROWSE

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[VIEW](#) [DOWNLOAD](#) [METADATA](#) [DETAIL](#)

Dimensions: 45 × 20 × 1080 × 1280

Slice (Optional) [EXAMPLES](#)

If blank, access entire array

Format *
CSV

[DOWNLOAD](#) [LINK](#) [OPEN](#)

Link
<https://tiled-demo.blueskyproject.io/a>

Tool #3: Collaborate



[Home](#) [About Lightsources.org](#) [75 Years of Synchrotron Science](#) [For Users](#) [Careers](#) [Light sources of the world](#)

Light sources of the world

There are more than 50 light sources in the world (operational, or under construction). This page lists all the members of the lightsources.org collaboration.



Map Satellite

Google

Keyboard shortcuts Map data ©2023 Imagery ©2023 NASA, TerraMetrics Terms of Use

Orange pins on the map represent members of the lightsources.org collaboration.
If your facility is not listed below and you wish to join our collaboration, please contact our [project manager](#).

We Work Openly

- Much of our work is public on the Internet.
 - Only site-specific details and credentials are internal. (They are boring anyway.)
- People can and do reuse and repurpose our work for free.
- We benefit from...
 - Quickly **discovering connections** to related efforts
 - Collecting **ideas** for capabilities or approaches
 - Hearing about **problems** we may not have run into yet
 - New capabilities **contributed back to us** by collaborators



99-80
September 8, 1999

For more information, contact:

**Diane Greenberg, (631)344-2347, greenb@bnl.gov,
or Mona S. Rowe, (631) 344-5056, mrowe@bnl.gov**

Brookhaven Lab to Hold "Open Source/Open Science" Conference, Oct. 2

UPTON, NY - The U.S. Department of Energy's Brookhaven National Laboratory will host a conference titled "Open Source/Open Science" on Saturday, October 2, from 8 a.m. to 5 p.m. in the Laboratory's Berkner Hall. The public is invited to the conference, but pre-registration is required. The fee for the conference is \$25, which includes lunch.

Open Source software is free software, typically released over the Internet so that a broad audience can evaluate and test it. The conference will bring together scientists and developers to exchange ideas about the use of Open Source software in scientific research at Brookhaven, as well as at other laboratories and universities. Talks by invited speakers, posters, demonstrations and vendor exhibits will be featured at the conference. Also offered will be tours of some of Brookhaven Lab's facilities in which Open Source software is used.

Among the talks and speakers featured at the conference will be:

- "What is Open Source?" Bruce Perens, the principal author of the Open Source definition
- "The Open Science Project," by Dan Gezelter from openscience.org
- "Open Source Computing for BNL's Relativistic Heavy Ion Collider," Tom Throwe, Brookhaven Lab
- "Open Visualization Data Explorer," Bill Horn, IBM
- "Open GL and GLX: High Performance 3D Graphics for Linux," Jon Leech, SGI
- "ACEDB: An Open Source Object-Oriented Database," Lincoln Stein, Cold Spring Harbor Laboratory
- "Open Source in Medical Imaging," Bill Rooney, Brookhaven Lab
- "Using Beowulf for Macromolecular Crystallography at the National Synchrotron Light Source," Malcolm Capel, Brookhaven Lab

Also, a panel of experts will discuss the issues that must be overcome by federal facilities in order to use and contribute to Open Source technologies.

To register for the conference, go to the Web site <http://openscience.bnl.gov>, and follow the registration instructions. The registration deadline is September 26. For more information, call (516)344-3582 or (516)344-5682.

Brookhaven National Laboratory is located on William Floyd Parkway (County Road 46), one-and-a-half miles north of Exit 68 on the Long Island Expressway.

The U.S. Department of Energy's Brookhaven National Laboratory creates and operates major facilities available to university, industrial and government personnel for basic and applied research in the physical, biomedical and environmental sciences, and in selected energy technologies. The Laboratory is operated by Brookhaven Science Associates, a not-for-profit research management company, under contract with the U.S. Department of Energy.

* * *

Collaboration is easier than it used to be

The screenshot shows the GitHub interface for the 'bluesky' repository. At the top, there's a navigation bar with links for Code, Issues (133), Pull requests (42), Discussions, Actions, Projects, Wiki, Security, Insights, and Settings. Below this, the repository name 'bluesky' is displayed with a 'Public' label. To the right, there are buttons for 'Edit Pins', 'Unwatch' (26), 'Fork' (76), and 'Star' (117). The main content area shows a list of files and folders with their commit history. A pull request by 'evalott100' is highlighted at the top. On the right, the 'About' section provides a description of the project as 'experiment orchestration and data acquisition' and includes links to the project website and tags like 'python', 'dataacquisition', and 'bluesky'.

bluesky / bluesky

Code Issues 133 Pull requests 42 Discussions Actions Projects Wiki Security Insights Settings

bluesky Public

Edit Pins Unwatch 26 Fork 76 Star 117

master 15 branches 73 tags

Go to file Add file Code

evalott100 Merge pull request #1616 from evalott100/pack_stream_resource_... 6e3c472 4 days ago 4,594 commits

.github	Merge branch 'master' into dep-python-version	last year
bluesky	Fixed error	4 days ago
continuous_integration/scripts	Switch to GH Actions.	3 years ago
docs	DOC the value is recorded in the stop doc as <code>abort</code> , make it match h...	2 weeks ago
fuzz	FMT: renamed variable in commented code	3 years ago
.codecov.yml	CI: Add codecov config.	4 years ago
.coveragerc	CI: tweak coverage settings	6 years ago

About

experiment orchestration and data acquisition

blueskyproject.io/bluesky/

python dataacquisition bluesky

Readme

BSD-3-Clause license

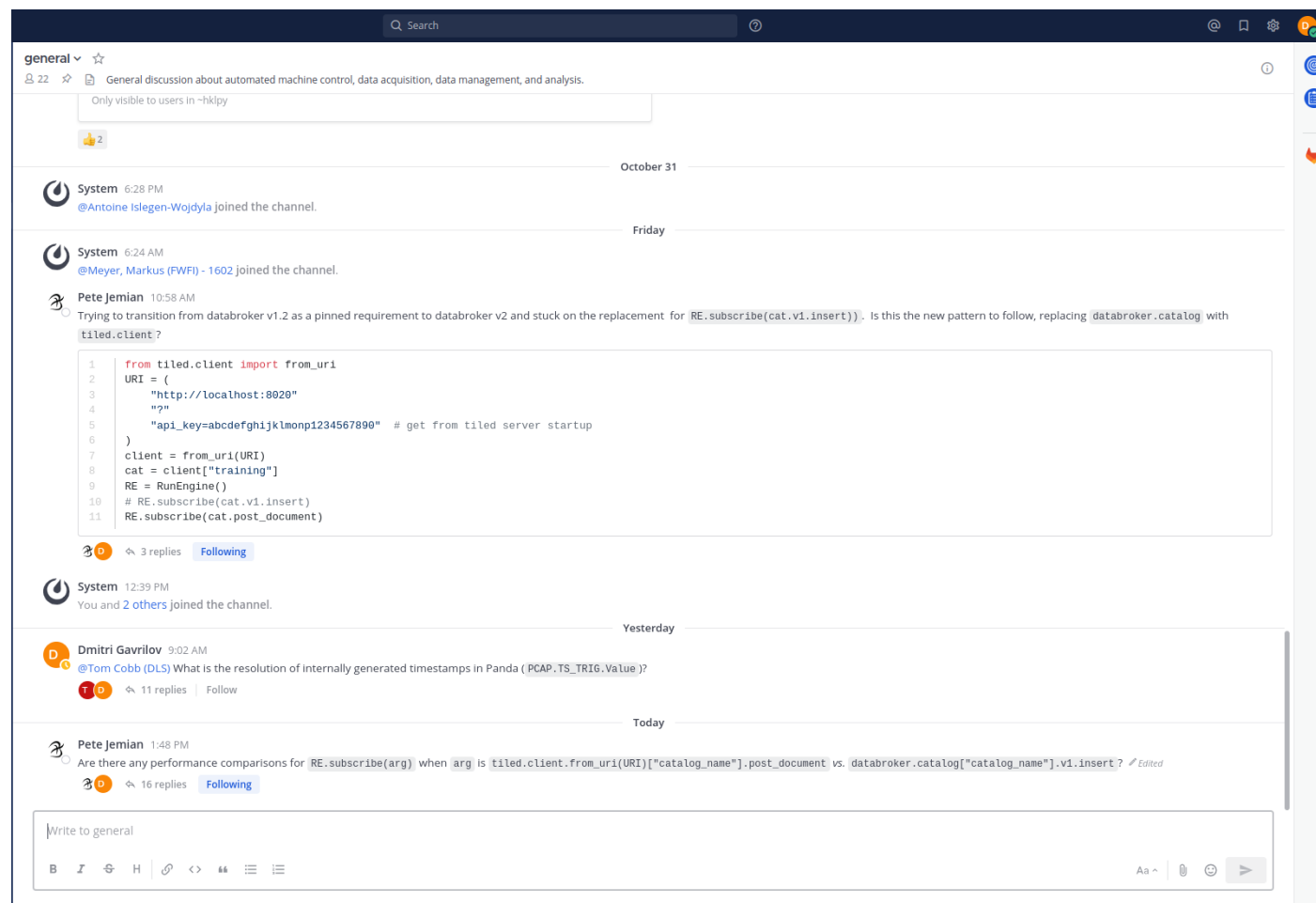
Activity

117 stars

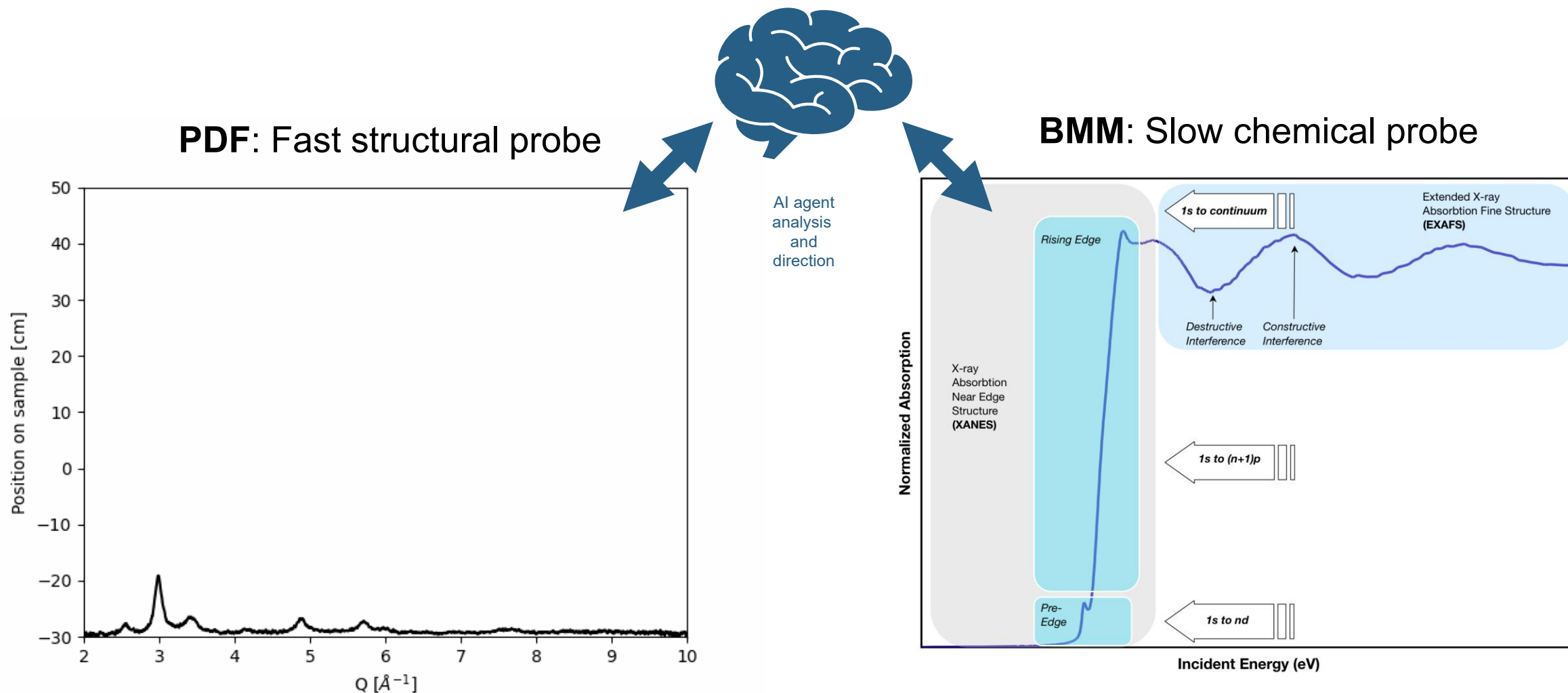
26 watching

76 forks

24 / 7 worldwide community chat



Example of What This Can Enable: Two Beamlines Sharing One "Brain"



We're still going to need *some* duct tape

To manage fast-changing requirements we use:

1. Sound planning and engineering
2. Abstraction
3. Collaboration



Part 2: The *Bluesky* Approach

What is Bluesky?

- An **open-source** Python-based toolbox for scientific data acquisition
- A bridge between the Scientific Python ecosystem and facility-scale or bench-scale experimental data acquisition setups
- Mini "ecosystem" of modular Python libraries
 - Python **hardware abstraction** above EPICS (or Tango, or custom)
 - Experiment **sequencing engine**
 - Portable **experiment logic** ("plans")
 - Streaming-first, metadata-rich **data model, unopinionated about data formats**
 - A growing number of libraries, services, and applications built on these

What do we mean by "open source"?

1. Publicly visible source code
2. Licensed for reuse with an OSI-approved license
3. Accepting contributions
4. Open development
5. Open decision making
6. Multi-institution engagement

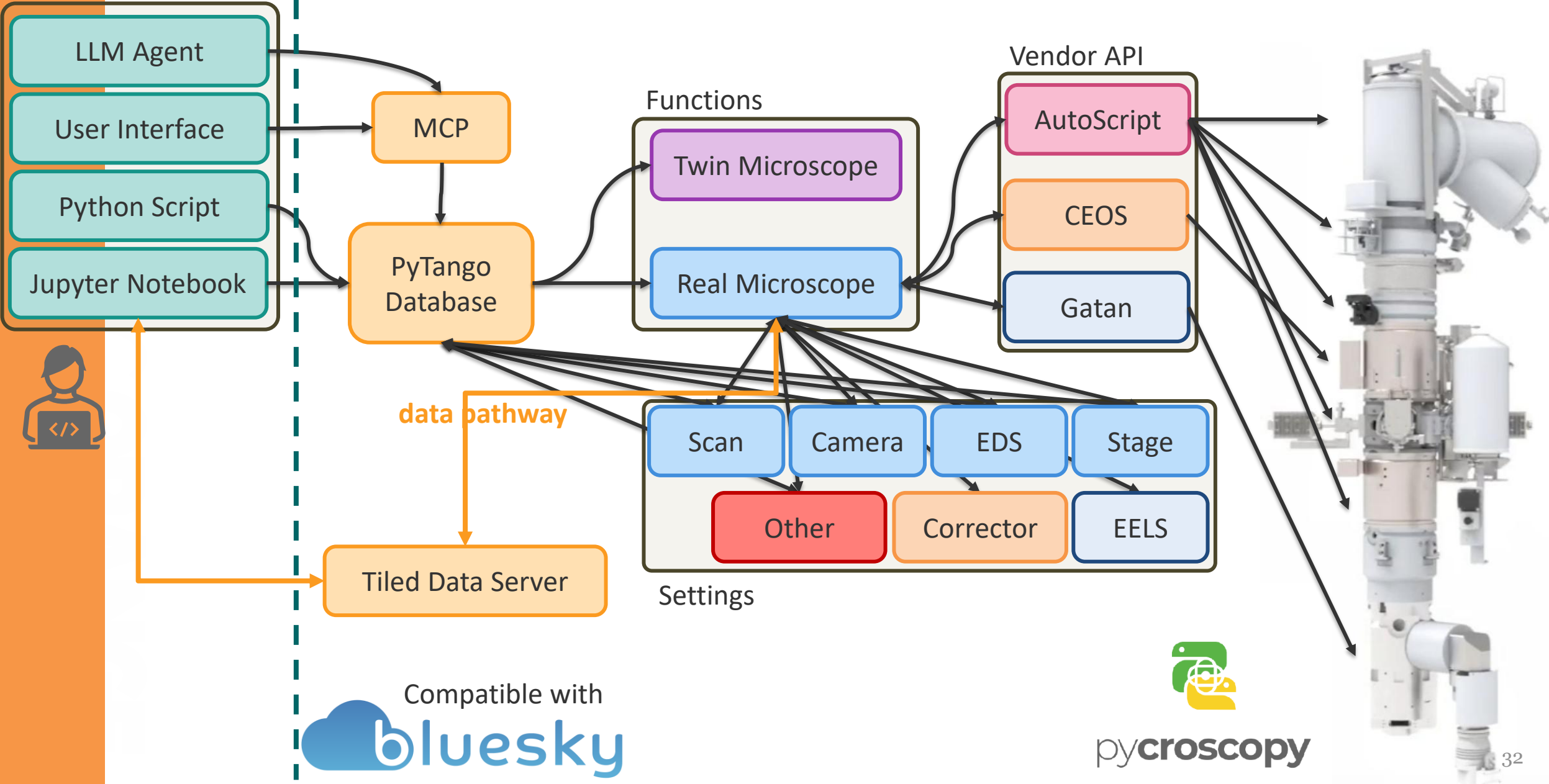


from Matt Rocklin's post
<https://www.coiled.io/blog/stages-of-openness>

Who uses Bluesky?

- Significant adoption at...
 - 5 DOE light sources
 - Australian Synchrotron
 - BESSY-II
 - Pohang Light Source
- Diamond Light Source
 - Multiple FTEs making major contributors to Bluesky for over 2 years
 - Bluesky will underpin all scanning for the new beamlines delivered by Diamond-II upgrade
 - Scope for adoption at existing beamlines is under discussion
- Growing adoption at Canadian Light Source (new team)
- Some use at CHESS, PSI (Swiss Light Source)
- Various academic labs and other facilities
- Starting in EM: <https://github.com/pycroscopy/asyncroscopy>

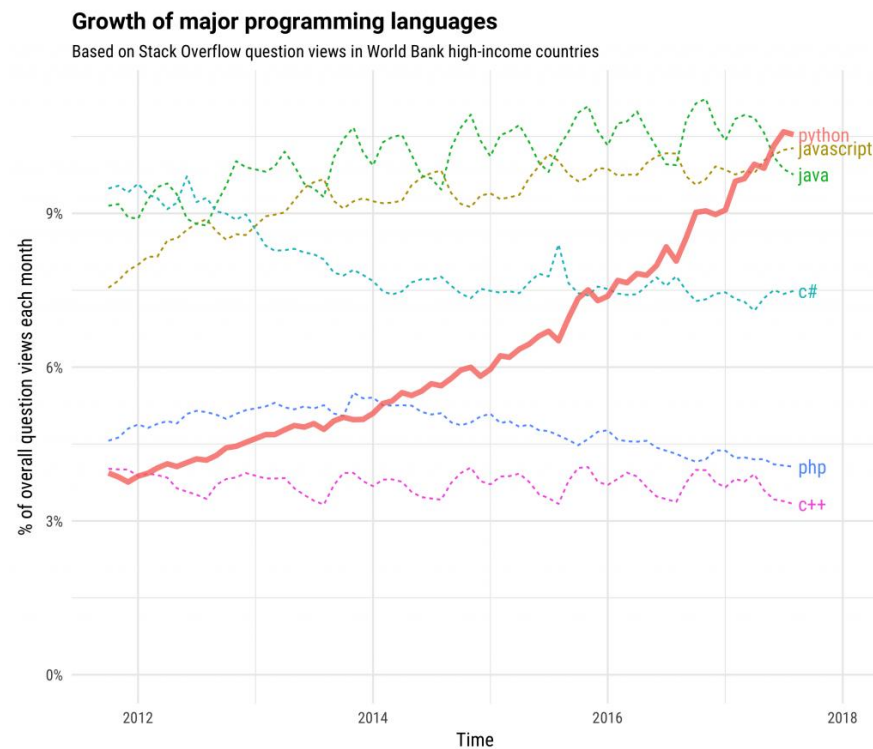




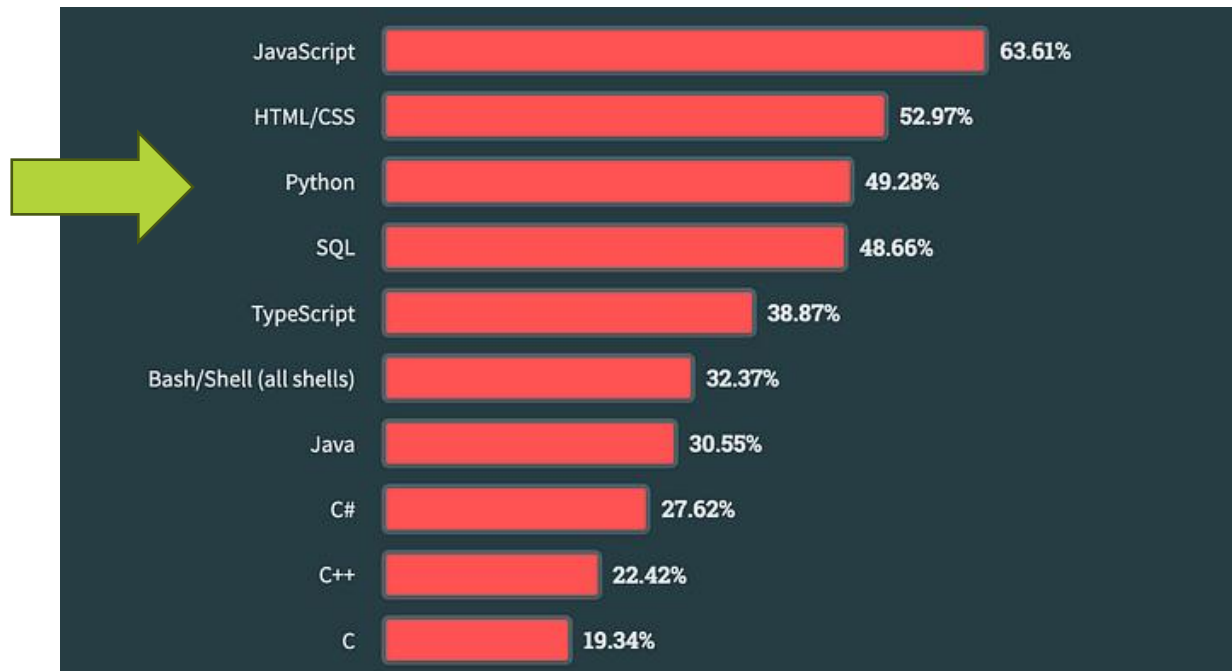
How does collaboration work at scale?

- Adoption can be incremental: take some pieces, leave or customize others.
- This is not an all-or-nothing framework that you have to buy into; it's a **mini-ecosystem of co-developed but individually useful tools** that you can build on.
- It's all in **Python**. Some beamline staff and partner users have built on it.
- The scientific Python community is an example of how distributed, loose collaboration can work well.

Why Python? It is a widely-known language with a growing, friendly community



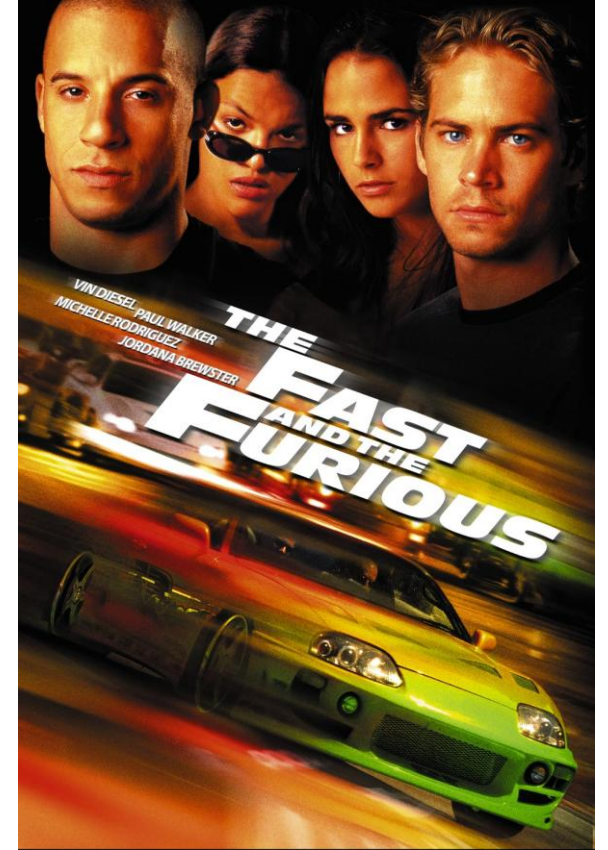
2012-2018: Growth of Python for Data Science



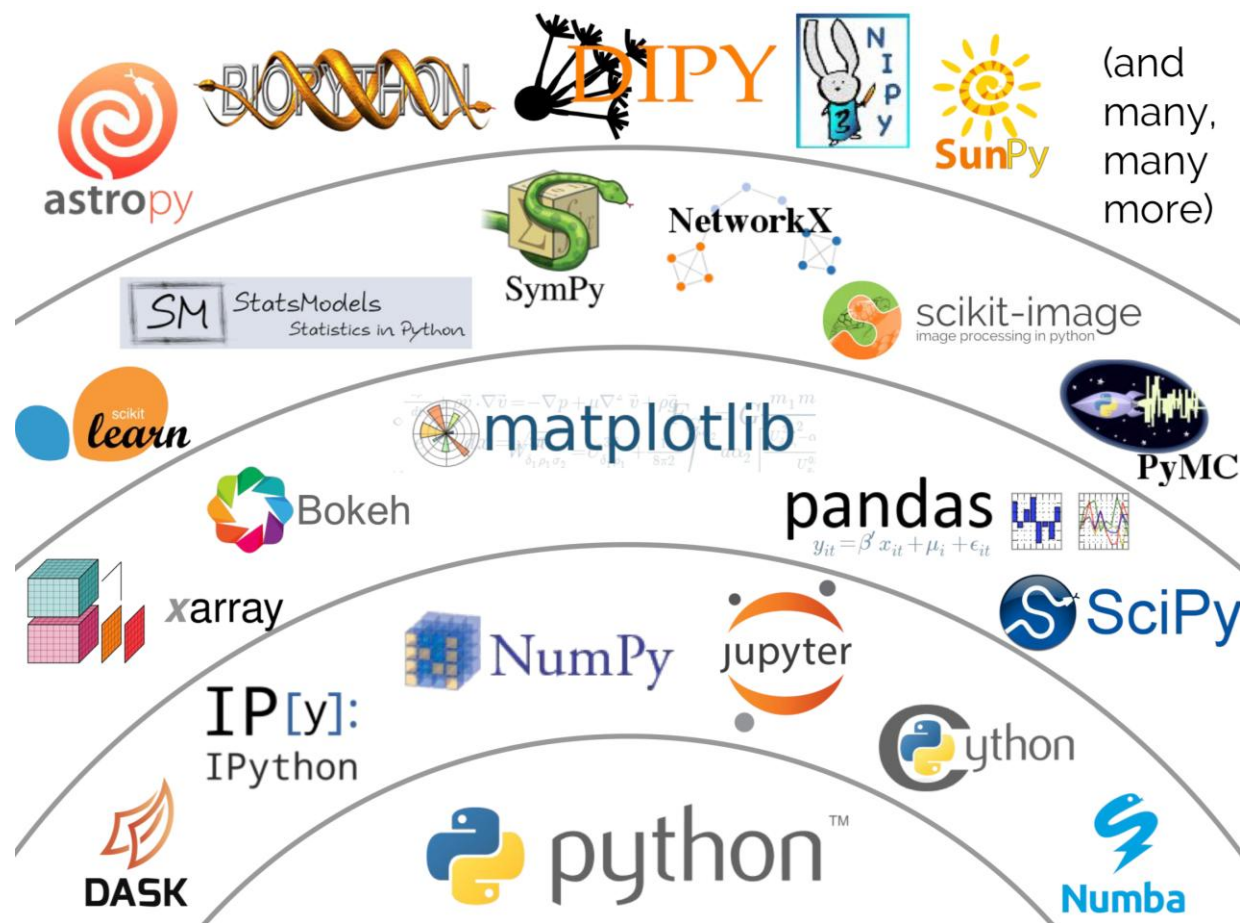
2023: Python "takes center stage" as a friendly and versatile programming language, now often taught in introductory scientific computing courses

Python is not the bottleneck for speed

- Python as the "glue".
- Expensive numerical computations happen in C or Fortran, wrapped by Python.
- Very often, the actual bottlenecks are waiting for hardware to move, or for data to transfer from disk or over the network.



Scientific Python is a *layered* ecosystem



Some Technical Goals of Bluesky (1 / 2)

- Working backward the needs of data analysis, capture all the important context.
- Be unopinionated data *formats*; focus on data *structures*.
- Support streaming.
- Handle asynchronous data streams.
- Support multi-modal: simultaneous, cross-instrument, cross-facility experiments.

Some Technical Goals of Bluesky (2 / 2)

- Form a bridge to the successful scientific Python ecosystem.
- Be generic across science domains.
- Scale up.
- Scale down.

Some Sociological Goals of Bluesky

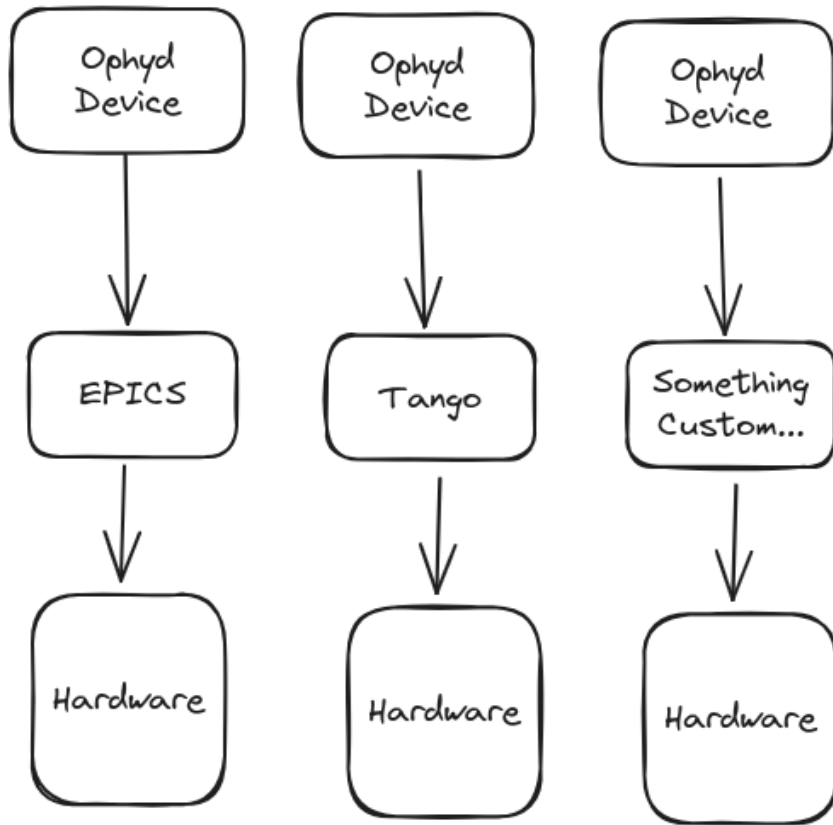
- Overcome "not-invented-here"-ism.
- Make components with well-defined boundaries, *co-developed but individually useful*, which can be adopted piecemeal by groups and facilities.
- Enable code to be reused in ways unforeseen by the original authors.
- Enable beamlines to present their users with a fine-tuned, polished user experience.

Share the toolbox



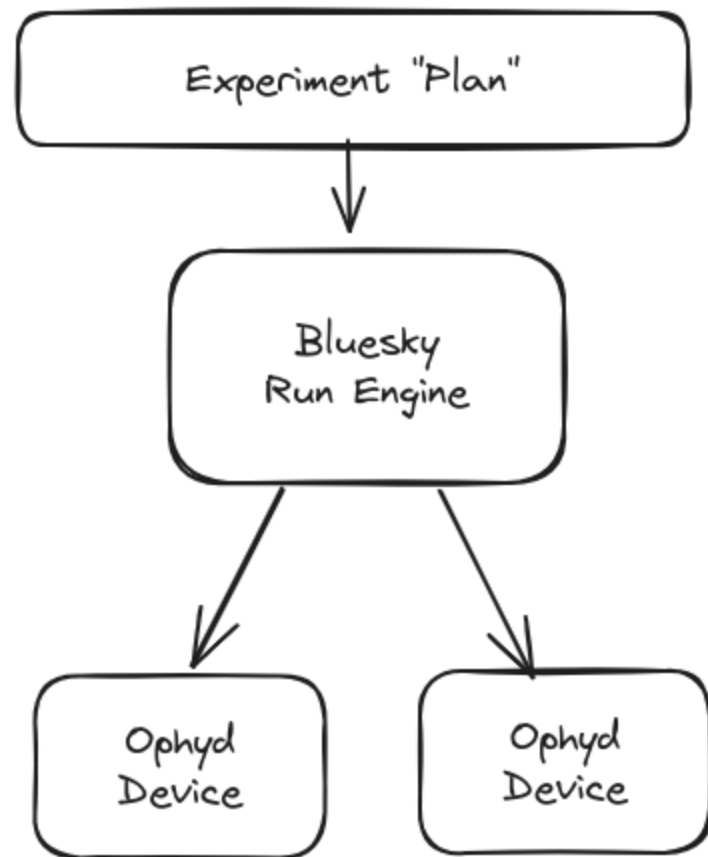
Part 3: *Bluesky* Architecture

Ophyd is a Python hardware abstraction



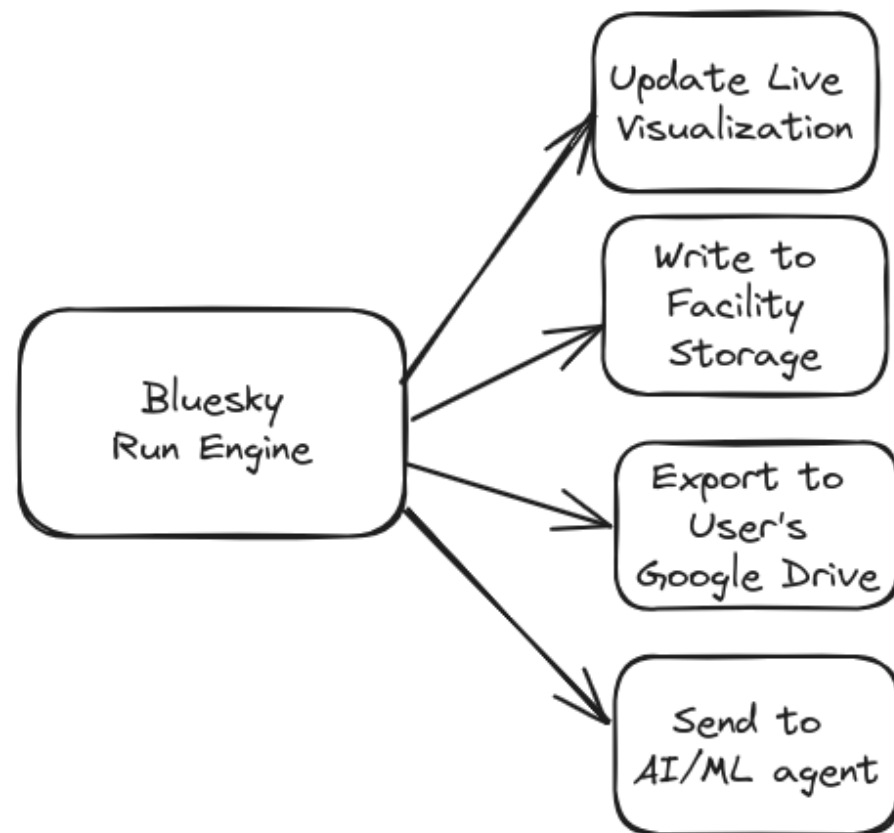
Provide high-level verbs like "trigger", "read", "set", "stop", ... over top of lower-level commands in EPICS or other control systems

Bluesky *Run Engine* orchestrates experiments



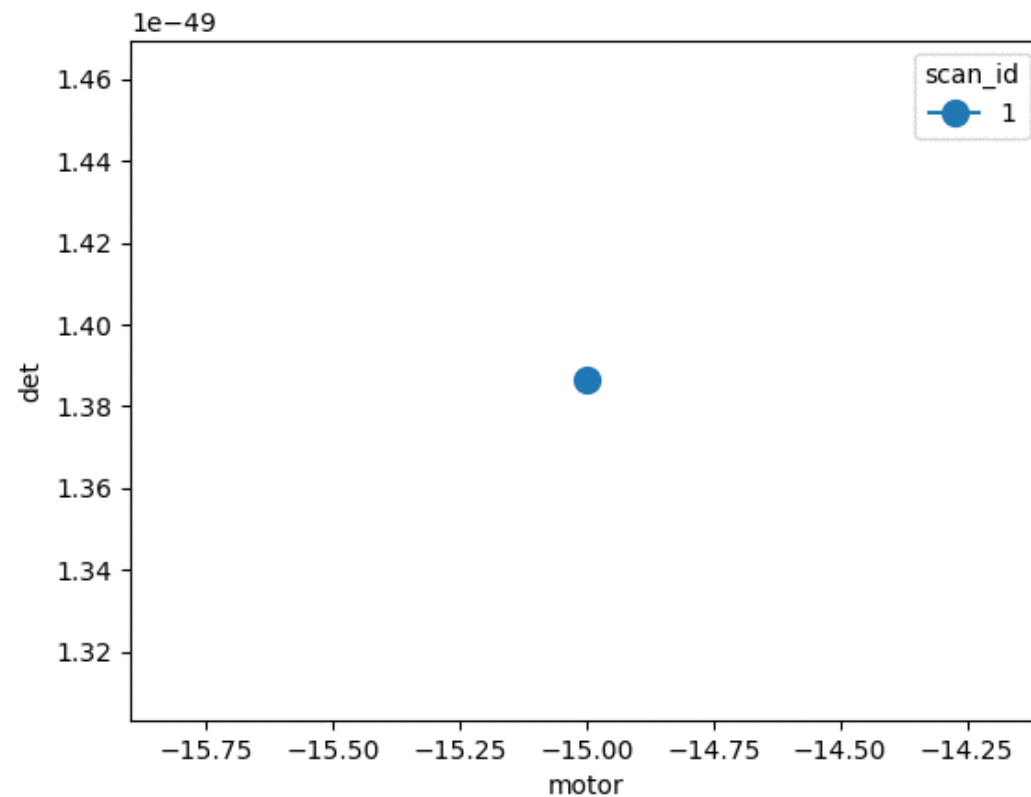
Execute portable "plans" on given hardware, handling feedback (e.g. progress bars), interruptions, and failure modes.

Bluesky *Run Engine* broadcasts metadata and data to consumers



Stream metadata and data to a configurable set of consumers.

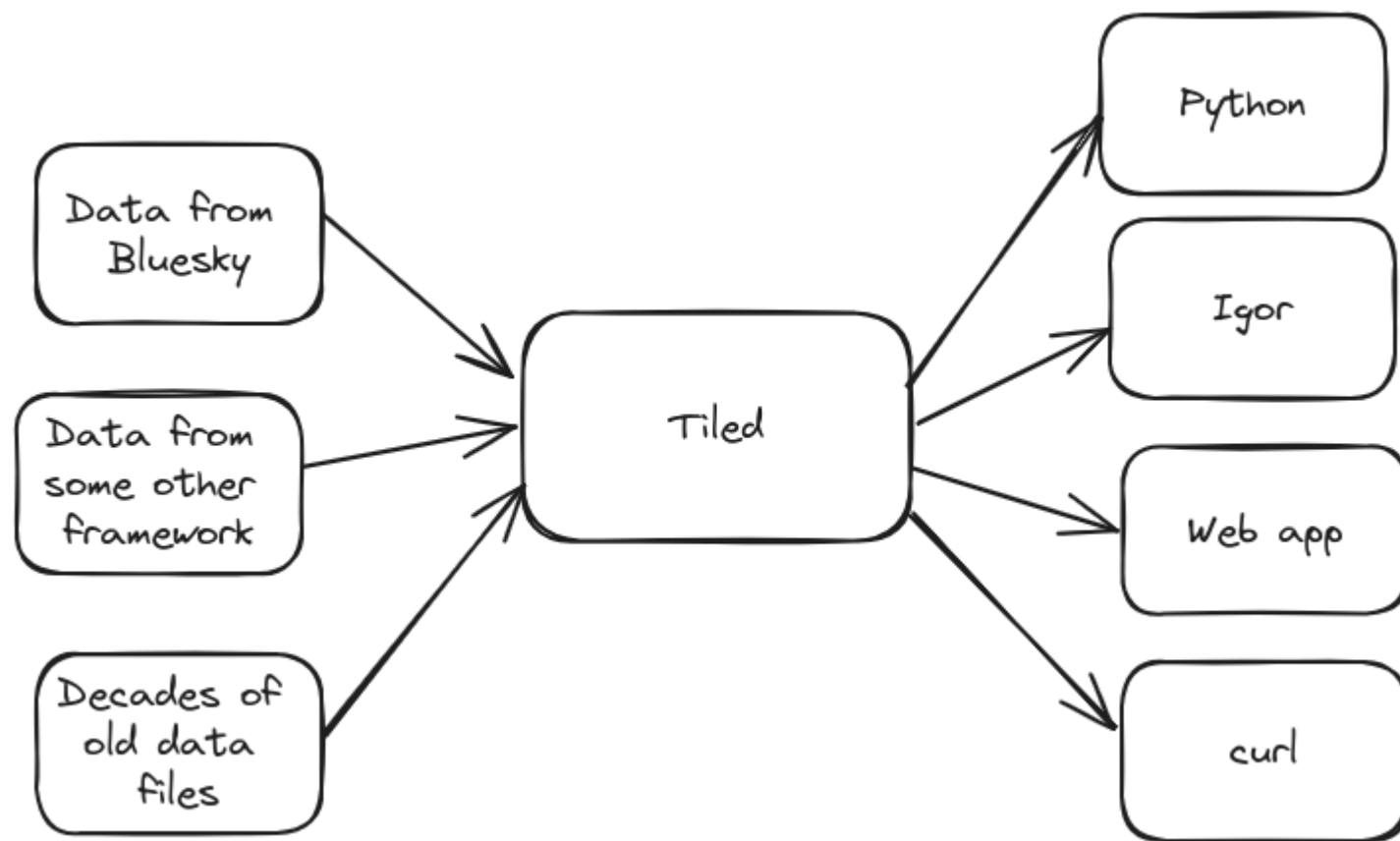
Example: Adaptive Scan with Live Plot



Data and Metadata are emitted in a streaming fashion in Documents

- Documents are JSON, a machine- and human-readable standard
- Follows a lightweight schema with open-ended space for scientific metadata
- Small data can go directly in the document; large data is referenced (e.g. a filepath and location in file)
- How data is *chunked* into these documents matters.
 - To make things go fast, careful choices need to be made here.
 - New work, in collaboration with Diamond Light Source, is focused on high-performance (fast) configurations for fly-scanning.

Tiled provides convenient data access

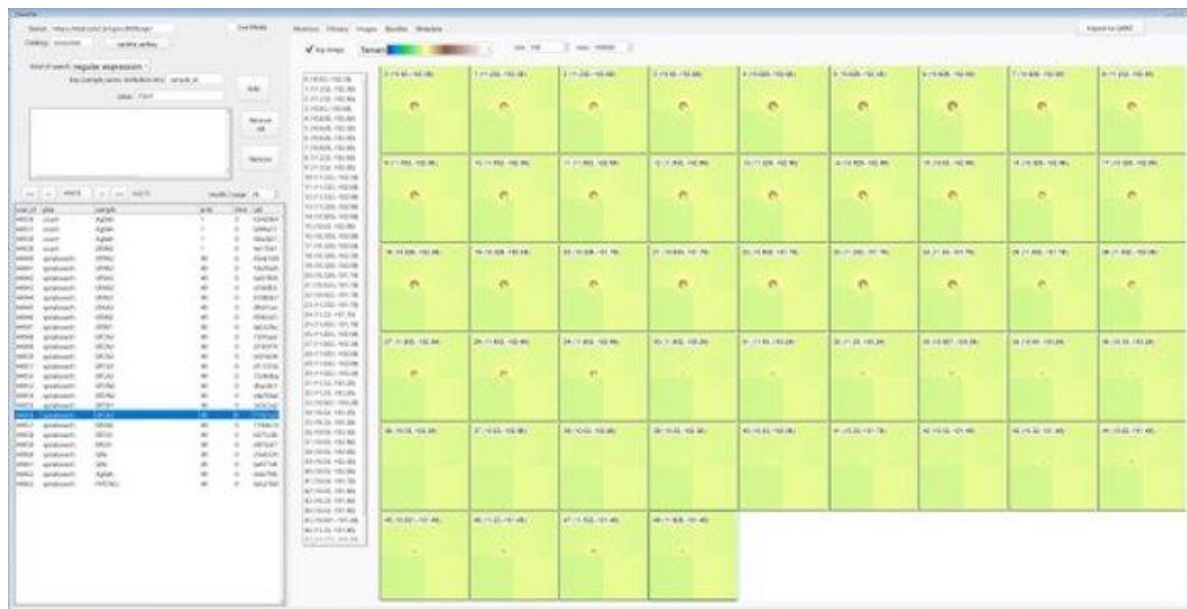


Make data searchable, sliceable, and accessible in many formats, regardless of how it is stored

Tiled generalizes an older project, *Databroker*:

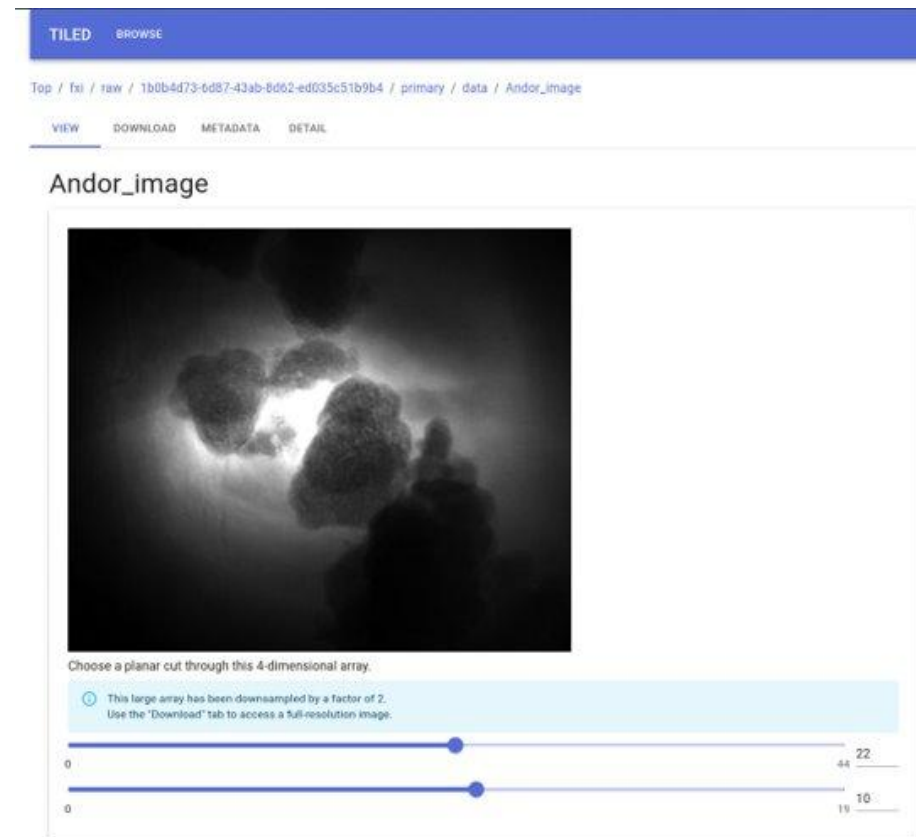
- Not just Python
- Not just data that came from Bluesky

Slice and/or transcode remotely

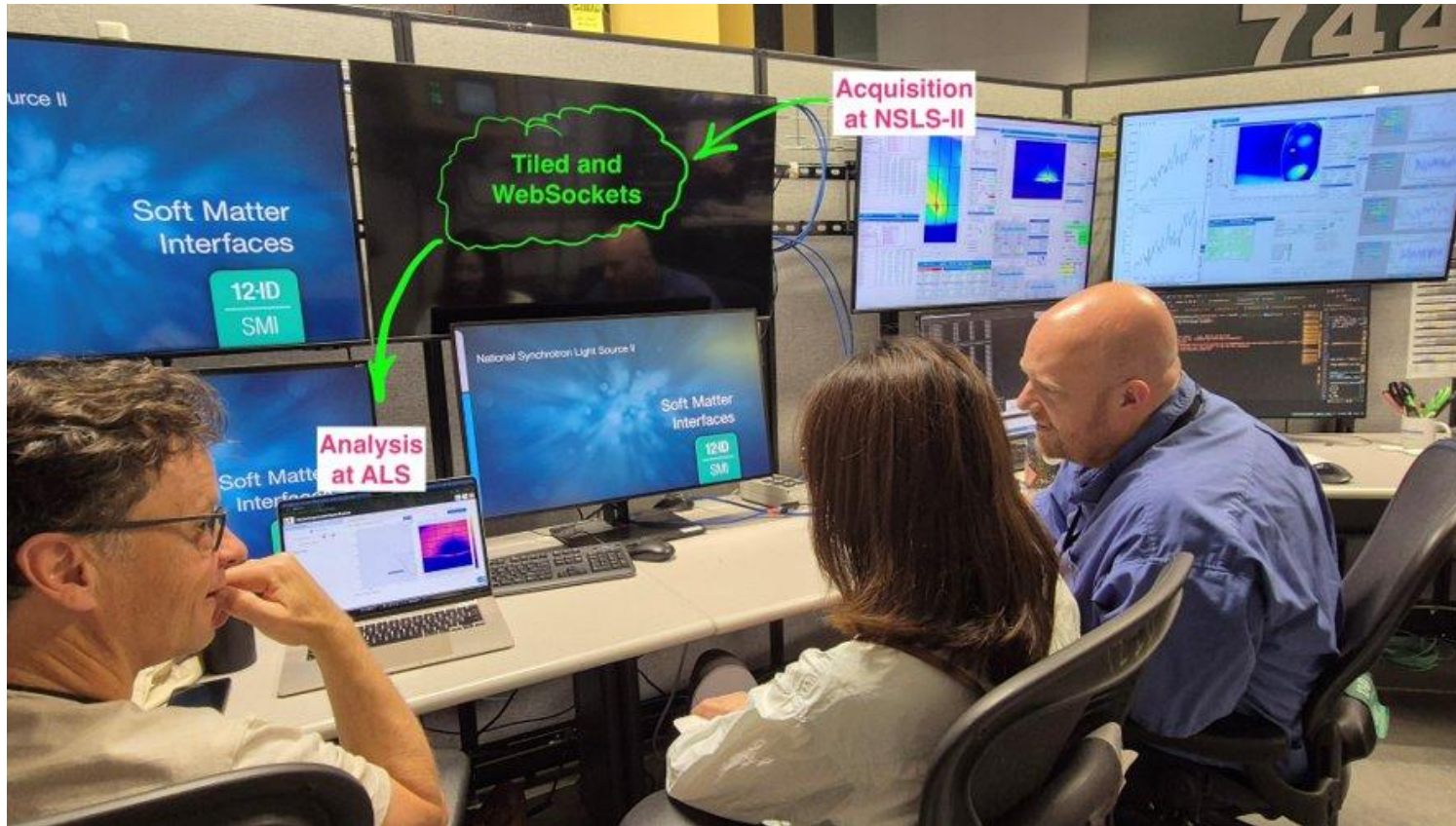


In Python: `x[:5, -10:, ::2]`

In HTTP: `https://...?slice=:5,-10:,:2`



Stream live



Typically, data is loaded upon request. Streaming flips this around: when streaming, Tiled **pushes** data to us as soon as it is written.

Recent data can be replayed so late subscribers can **catch up**.

Technical note:
Some data is streamed via the WebSocket connection before it is even saved to disk, which minimizes latency. Redis is used to cache recent data.

Wrapping up...

- *Bluesky* is a multi-facility open-source collaboration.
- It is a toolbox, not a ready-to-use application.
- The core Python libraries are well established and have been stable since 2017.



Areas of Active Development (1 / 1)

- **Graphical Applications**
 - Integrations with applications our users already use
 - New applications
- **Services**
 - Run Engine as a service (Queueserver, Blue API)
 - Data Access (Tiled)



Areas of Active Development (2 / 2)

- Lean into **Autonomous Experiments** and Adaptive Control Logic
 - Bluesky has always supported this
 - BLOP for optimization
- Enrich **Fly-scanning**
 - Led by Diamond, the collaboration has built fly-scanning components that do more right out of the box
- Remote experiment control
 - Queueserver (stable backend) and ophyd-as-a-service (early prototype)
 - Finch (react components to build web frontends)

Questions & Discussion

Extra Slides...

Minimum Viable Governance

- Maintainers: per repo, make day-to-day decisions and set processes as appropriate to the repo
- Technical Steering Committee: arbitrate when maintainers cannot reach rough consensus
- Project Advisory Board: management-level stakeholders, oversee big-picture priorities

